

**POST GRADUATE DIPLOMA IN
COMPUTER APPLICATIONS
(PGDCA-NEW)**

Term-End Examination

December, 2022

MCS-202 : COMPUTER ORGANISATION

Time : 3 Hours

Maximum Marks : 100

Note : *Question Number 1 is compulsory and carries 40 marks. Attempt any **three** questions from Question No. 2 to Question No. 5.*

1. (a) Perform the following operations using signed 2's complement notation using 8 bit registers. (All the numbers given below are in decimal notation). Also indicate overflow, if any :

6

- (i) Add (-59) with (-69)

- (ii) Subtract (-69) from (-59)
- (iii) Add $+59$ with $+69$
- (iv) Add -75 and 35
- (b) Simplify the following Boolean expressions :
- 4
- (i) $(A + B)' + (A' + B)'$
- (ii) $((AB)' + (A'B)')'$
- (c) Explain how data is organised in a magnetic disk with the help of a diagram. Also differentiate between Constant Linear Velocity (CLV) and Constant Angular Velocity (CAV) disks. 6
- (d) Explain the programmed I/O with the help of a diagram. 4
- (e) Explain different components of an instruction of a computer system. Explain the subroutine call and return instructions with the help of sequence of operations that would be required to implement these instructions. 6

- (f) Explain the organisation of control memory with the help of a diagram. 5
- (g) Explain the PUSH instruction of 8086 microprocessor with the help of an example. 3
- (h) Write a program using 8086 assembly language that converts a 2-digit ASCII number stored in memory to equivalent binary number. 6
2. (a) Explain the Von Neumann architecture with the help of a diagram. Differentiate the features of Von Neumann architecture with the Harvard architecture. 7
- (b) Simplify the following Boolean function Product of Sum (SOP) form using K-map. Draw the logic diagram of the resulting function : 6
- $$F(A, B, C, D) = \Sigma (0, 4, 5, 6, 8, 14, 15)$$
- (c) Explain the J-K flip-flop with the help of a block diagram and characteristic table. Make the excitation table of J-K flip-flop.

3. (a) What is cache memory ? Why is it needed ?

If hit ratio of a cache memory is 80% and cache access time and memory access time for that machine are 5 nanoseconds and 100 nanoseconds respectively, then find the effective memory access time for this system. 5

- (b) Explain the 'set-associative mapping' scheme of main memory to cache addresses with the help of a diagram showing how a main memory address can be mapped to cache address. 7

- (c) Explain the features of the following I/O devices or components : 8

- (i) Scanner
- (ii) Video Memory
- (iii) Digital Camera
- (iv) LEDs

4. (a) Define the term micro-operation. Explain the following micro-operations with an example of each : 7
- (i) Register-transfer micro-operation
 - (ii) Arithmetic micro-operation
 - (iii) Logic micro-operation
 - (iv) Shift micro-operation
- (b) Differentiate between hardwired control unit and micro-programmed control unit. Explain the functioning of Wilke's control unit with the help of a diagram. 7
- (c) List the features of RISC architecture. Explain the instruction pipelining in the context of a RISC architecture. 6
5. (a) Explain the use of segment registers in 8086 microprocessor with the help of examples. 5
- (b) Explain the use of INT 21h in the context of 8086 microprocessor for performing input and output operations with *one* example of each. 5

- (c) Write a program using 8086 assembly language that finds the highest value in an array of 5 unsigned byte values stored in the memory. 7
- (d) Explain the concept of vector processing with the help of an example. 3

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Note : *Question Number 1 is compulsory and carries 40 marks. Attempt any **three** questions from Question No. 2 to Question No. 5.*

1. (a) Perform the following operations using signed 2's complement notation of 8-bits, (including the sign bit). You must indicate overflow or no-overflow and the reasons thereof :
- 6
- (i) Add + 69 and + 59
- (ii) Subtract 90 from – 38
- (iii) Add – 76 and + 86

- (b) Simplify the following function using Karnaugh's map : 5

$$F(A, B, C, D) = \Sigma(2, 3, 6, 7, 9, 10, 11, 12, 15)$$

Draw the logic diagram of the simplified function using AND, OR and NOT gates.

- (c) What is the need of Cache memory ? Explain the associative mapping scheme of Cache memory with the help of an example. 5
- (d) Explain with the help of a flowchart how Interrupt-driven Input/Output technique is used to perform input function. 5
- (e) Explain the following addressing modes with the help of *one* example each : 6
- (i) Register indirect addressing
 - (ii) Base register addressing
 - (iii) Direct addressing
- (f) What is instruction pipelining ? Explain with the help of a diagram. 4
- (g) What are the segment registers in 8086 microprocessor ? Explain their uses with the help of an example. 4

- (h) Write a program in 8080 assembly language that finds the largest of two byte numbers stored in the memory. The program then subtracts smaller number from the larger number. The result is left in AX register. 5
2. (a) Explain the Von-Neumann architecture. How is it different than the Harward architecture ? 5
- (b) Explain different components of a floating point number. How are they different than the fixed point numbers ? 5
- (c) Draw the S-R latch using NOR gates and explain its functioning. Also make the characteristic table of S-R flip-flop. 5
- (d) Draw the truth table and block diagram for synchronous counter. Explain its functioning. 5
3. (a) Explain the term access time in the context of a hard disk. Also differentiate between CLV and CAV in the context of secondary storage technologies. 5
- (b) Explain the set-associative cache mapping scheme. 6
- (c) What is DMA ? Explain the structure of DMA module with the help of a block diagram. 5

- (d) Explain the following terms in the context of Input/Output technologies : 4
 - (i) Colour depth
 - (ii) Non-impact printers
- 4. (a) Explain the steps of subroutine call and return instructions with the help of an example. Assume suitable register set. 4
- (b) What is a micro-operation ? Explain register-transfer, arithmetic and logic micro-operations with the help of an example. 7
- (c) Explain the features of RISC processors. 4
- (d) Explain the operations of micro-programmed control unit with the help of a block diagram. 5
- 5. (a) Explain the following in the context of 8086 microprocessor : 15
 - (i) Bus interface unit
 - (ii) Tools for assembly language programming
 - (iii) Interrupt processing in 8086 micro-processor
- (b) Write a program in 8086 assembly language that converts an ASCII digit stored in a memory location into its equivalent binary. 5

**POST GRADUATE DIPLOMA IN COMPUTER
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Term-End Examination

June, 2022

MCS-202 : COMPUTER ORGANISATION

Time : 3 hours

Maximum Marks : 100

(Weightage : 70%)

Note : *Question number 1 is **compulsory** and carries 40 marks. Attempt any **three** questions from questions number 2 to 5.*

1. (a) Convert the following : **6**

(i) Decimal number 249·125 to binary, octal and hexadecimal

(ii) Hexadecimal $(FA12)_h$ into octal, binary and decimal

(b) Use K-map to find the optimal function for the function

$$F(A, B, C, D) = \sum (0, 4, 8, 11, 12, 13).$$

Also draw the logic diagram of resultant function using AND, OR, NOT Gates. **5**

- (c) Draw and explain the logic diagram of a RAM cell. 5

- (d) A main memory has the size of 2 MB. The word size of the main memory is 16 bits and 1 block of main memory word/cache memory is of 64 bits. The size of the cache memory is 64 KB. Determine the following : $1+1+3=5$

- (i) Size of the address of main memory.
- (ii) The number of cache lines.
- (iii) Assuming that a main memory to cache direct address mapping is used, explain how main memory address will be mapped to cache memory address.

- (e) Consider the following state of registers and main memory (All values are in Hexadecimal) : 4

Address	Main Memory
4F11	LDA
4F12	4F14
4F13	2567
4F14	6782
4F15	
4F16	4F00
4F17	4F14

2511 AC

IR

4F11 PC

0002 XR

0003 BR

Assuming that the load instruction is stored in address 4F11 and 4F12, where the location 4F12 contains the address field of the load instruction. Calculate the value of operand, that would be loaded to AC, if load instruction uses the following addressing modes :

- (i) Memory Direct Addressing
 - (ii) Index Addressing
 - (iii) Base Addressing
 - (iv) Immediate Addressing
- (f) Explain the sequence of micro-operations that would be required to fetch an instruction stored in a memory location to IP register. You may assume that the machine uses MAR $\rightarrow$ register to store memory address, R1 $\rightarrow$ register which stores data received from memory or data to be transferred to memory; AC $\rightarrow$ register which is used to perform computation, IP $\rightarrow$ register that stores the instruction, which is fetched and PC $\rightarrow$ register which stores the instruction to be fetched. Make suitable assumptions. 5
- (g) For the following values of various registers in an 8086 microprocessor, compute the physical memory address : 4
- (i) CS : IP $\equiv$ 0FABh : 01FAh
 - (ii) SS : SP $\equiv$ 1F0Ah : 0267h

- (h) Write a program in 8086 assembly language that interchanges the value stored in two memory locations having the name FIRST and SECOND. Explain the logic of the program. 6
2. (a) Simplify the following Boolean functions : 4
- (i) $((A + B')')' + A \cdot B'$
- (ii) $(A + B)' + (A \cdot B)' + (A' \cdot B)'$
- (b) Explain the process of error detection and correction using parity bit. 5
- (c) Explain the floating point number representation with the help of an example. 5
- (d) Explain the working of a basic SR latch. Draw the characteristics table of SR flip-flop. Also make the excitation table of SR flip-flop. 6
3. (a) Explain the internal structure of a hard disk drive with the help of a diagram. 5
- (b) Explain the concept of memory hierarchy with the help of an example. 5
- (c) What is an Interrupt ? How can it be used for Input/Output ? Explain interrupt driven I/O with the help of an example. 5
- (d) Briefly explain any two types of printer technologies. 5

4. (a) Explain the following program control instructions with the help of an example : 2+2+1=5
- (i) Branch
 - (ii) CALL
 - (iii) Return
- (b) What is the need of an interrupt cycle during the instruction execution ? Explain the micro-operations of an interrupt cycle. Make suitable assumptions. 5
- (c) Explain the working of Wilkes control unit with the help of a diagram. 5
- (d) Explain the use of large register file with the help of an example. 5
5. (a) Explain any three shift instructions in 8086 microprocessor. 6
- (b) Differentiate between .com and .exe programs in the context of 8086 microprocessor. 4
- (c) Write a program using 8086 assembly language that adds four byte values stored in an array of size 4 in the memory. The result of addition is stored in the fifth and sixth location. 6
- (d) Explain the characteristics of the following : 4
- (i) Array processing
 - (ii) Multiprocessor system

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Note : *Question Number 1 is compulsory and carries 40 marks. Attempt any **three** questions from Question No. 2 to Question No. 5.*

1. (a) Perform the following operations using signed 2's complement notation of size 8 bits, which includes the sign bit. Also indicate overflow or no-overflow, giving reasons. Thereof : 6
(i) Add -39 and $+126$

(ii) Subtract – 38 from 90

(iii) Add –79 and –49

- (b) Simplify the following function using Karnaugh's map : 5

$$F(A, B, C, D) = \Sigma(5, 6, 8, 11, 13, 14, 15)$$

Draw the logic circuit for the simplified function.

- (c) Explain the Direct Cache mapping scheme with the help of an example. 5
- (d) What is an interrupt ? List all the steps that are required to be performed by hardware or software to process an interrupt. 5
- (e) Explain the steps required to process Branch (BR) instruction, subroutine call and return instruction of a computer which uses a stack for storing the return address. You may assume the necessary registers required for this purpose. 5

- (f) Explain the horizontal and vertical micro-instructions with the help of a suitable diagram. 5
- (g) Explain the following instructions of 8086 microprocessor with the help of an example : 4
- (i) SHL
- (ii) RCL
- (h) Write a 8086 assembly language program to find the largest value in an array consisting of 5 values. You may assume that these values are stored in the memory. The largest value may be kept in AX register. Make suitable assumptions. 5
2. (a) Perform the following conversions : 5
- (i) $(225)_{10}$ to hexadecimal
- (ii) $(AAA)_{16}$ to octal
- (iii) $(6761)_8$ to hexadecimal
- (iv) $(ABC)_{16}$ to decimal
- (v) $(225)_8$ to decimal

- (b) Explain the functions of the components of a central processing unit with the help of a diagram. 5
 - (c) Draw the truth table and logic diagram of a 4×1 multiplexer and explain its functioning. 5
 - (d) Draw the logic diagram of a ripple counter and explain its functioning. 5
3. (a) Explain the structure of magnetic disk with the help of a diagram. Also, explain the term constant angular velocity (CAV) in this context. 5
- (b) Explain the concept of memory interleaving with the help of an example. 4
 - (c) What is the role of Input/Output interfaces ? Explain. 5
 - (d) Explain the following terms in the context of Input/Output technologies : 6
 - (i) Refresh rate
 - (ii) Optical resolution
 - (iii) Impact printers

4. (a) Explain the following addressing schemes with the help of an example for each : 6
- (i) Immediate addressing
 - (ii) Direct addressing
 - (iii) Register addressing
- (b) Explain the microoperations required for Fetching an Instruction (FI). You may assume suitable registers for this. 5
- (c) Explain the working of a Wilke's control-unit with the help of a diagram. 5
- (d) Explain any *four* differences between the RISC and CISC machines. 4
5. (a) Compute the physical address in 8086 microprocessor for the following pairs : (all values are in hexadecimal) : 6
- (i) CS : IP $\equiv$ 10FFh : 0111h
 - (ii) DS : BX $\equiv$ 0322h : 20FFh
 - (iii) SS : SP $\equiv$ 0111h : 2222h

- (b) Explain the differences between .COM programs and .EXE programs, in the context of Assembly language, with the help of an example. 5
- (c) Write a program in 8086 assembly language to interchange the values of two byte locations in the memory. 5
- (d) Explain the concept of FAR and NEAR procedures in 8086 assembly language. 4

No. of Printed Pages : 5

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Note : *Question No. 1 is compulsory and carries 40 marks. Attempt any **three** questions from the remaining Question No. 2 to Question No. 5.*

1. (a) Explain Harvard architecture with the help of a diagram. 5

P. T. O.

- (b) Perform the following conversions : 5
- (i) Decimal $(873)_{10}$ to Binary
 - (ii) Decimal $(384)_{10}$ to Hexadecimal
 - (iii) Hexadecimal $(FAB)_h$ to Octal
 - (iv) ASCII string 'the escape sequence' to UTF8
 - (v) Octal $(765)_8$ to Decimal
- (c) Differentiate between CLV and CAV type of disk organizations. 5
- (d) Explain two-way set associative Cache mapping with a suitable example. 5
- (e) Differentiate between RISC processor and CISC processor. Give utility of each. 5
- (f) What is an interrupt ? Explain the use of interrupt in input/output with the help of an example. 5

- (g) List the various register categories involved with 8086 microprocessor. Also, give the role of each type of register category. 5
- (h) Compare direct addressing mode and indirect addressing mode. Give example for each. 5
2. (a) Simplify the function :
- $$F(A, B, C) = \Sigma (2, 3, 4, 5)$$
- using k-map. Also, draw the circuit using NAND gates. 5
- (b) Explain the functioning of 2×4 decoder. Draw its truth table and explain its logic diagram with the help of example input. 8
- (c) Explain the working of SR flip-flop with the help of a logic diagram and characteristic table. Also, explain the excitation table of SR flip-flop. 7

3. (a) Explain the concept of programmed I/O. How is Direct Memory Access (DMA) technique differ from programmed I/O technique ? Under what situation is it better to use DMA over programmed I/O ?

8

- (b) Define disk access time. Write the equation to compute disk access time. Also, discuss the meaning of all the parameters involved in the computation of disk access time. 7

- (c) What are Charge-Coupled Devices (CCDs) ? Briefly discuss the utility of CCDs. 5

4. (a) Explain the following addressing modes, with the help of a suitable diagram (showing content of sample memory and registers) :

10

- (i) Register Indirect Addressing
(ii) Relative Addressing

Note : Your diagram should include instructions, operands and addresses.

- (b) What is a micro-operation ? Explain register transfer micro-operation with a suitable example. 5
- (c) Write a program in 8086 assembly language, which converts a two digit ASCII number to a packed BCD number. Explain each step of the program. 5
5. Write short notes on the following : 5×4=20
- (i) Cache coherence
 - (ii) Parallelism in uniprocessor systems
 - (iii) .Com programs and .Exe programs
 - (iv) Wilkes control